Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The UDP protocol reveals that: The user's computer sent a DNS request using the UDP protocol to find the IP address of www.yummyrecipesforme.com. However, the DNS server could not process the request, which prevented the website from loading. This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message: The tcpdump log shows an ICMP error message stating "udp port 53 unreachable." This means the DNS request could not be delivered to the DNS service. The same error occurred multiple times when the computer tried to connect. The port noted in the error message is used for: Port 53 is used by the Domain Name System (DNS) to translate website names into IP addresses. DNS commonly uses UDP port 53 to process requests. The most likely issue is:The DNS service is unavailable or not accepting requests on UDP port 53. This prevents the computer from finding the website's IP address and accessing the webpage.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time incident occurred:The network traffic log shows the issue occurring at approximately 1:24 p.m. (13:24:32). The exact time customers first reported the problem was not provided. Explain how the IT team became aware of the incident:Several customers reported that they could not access www.yummyrecipesforme.com. They received a "destination port unreachable" error after waiting for the website to load. The cybersecurity team also experienced the same error when testing the website. Explain the actions taken by the IT department to investigate the incident:The cybersecurity team used tcpdump, a network traffic analyzer, to capture and examine packets sent between the user's computer and the DNS server. They checked the UDP requests and ICMP responses to understand why the website was not loading.

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.): The investigation showed that the computer at IP address 192.51.100.15 sent DNS requests to the DNS server at 203.0.113.2 using UDP port 53. Instead of receiving a DNS response, the computer received ICMP messages saying “udp port 53 unreachable.” This error appeared repeatedly, showing that the DNS requests were unsuccessful.Because the DNS lookup failed, the browser could not obtain the website’s IP address to establish a connection.

Note a likely cause of the incident:A possible cause is that the DNS service on the server is not running or is not listening on UDP port 53. Another possibility is that a firewall or network configuration is preventing DNS requests from reaching the service. The next step is for the security engineers to check the DNS server, confirm that the DNS service is running, and verify that UDP port 53 is properly configured and accessible.
Once the issue is fixed, the team should test the website again to confirm that DNS requests are successful and the webpage loads normally.